

AGENTIC HARNESS

The chassis that turns raw intelligence into a controllable system.

SUBJECT MODEL vs MODEL+HARNESS	METAPHOR FORMULA 1	SCALE 1:1	REV 1.1
DOC AH-001	SYSTEMS 05	STATUS RACE-READY	SHEET 1 / 1

01 / THE THESIS

The model is **not** the agent.

AGENT = MODEL + HARNESS

The model is the engine. The harness is everything around it that makes the engine **raceable** — structure, limits, sensors, control.

— WITHOUT HARNESS

Just an engine

- ◆ No chassis to mount it in
- ◆ Powerful, but uncontrollable
- ◆ Systemically non-deterministic

OUTCOME → UNPREDICTABLE

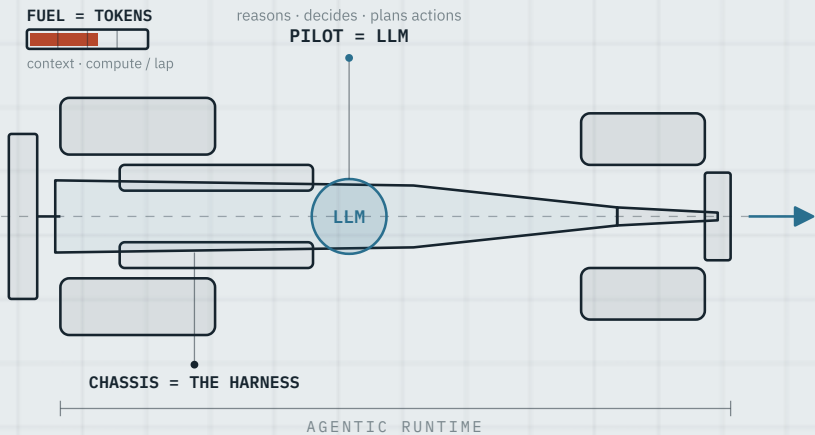
+ WITH HARNESS

A full race car

- A decision-making OS
- Constraints, observability, repeatability
- Reliable, safe, race-ready

OUTCOME → IN CONTROL

02 / THE MACHINE — PLAN VIEW



03 / INSIDE THE HARNESS — FIVE SYSTEMS

01

STRUCTURE

≈ THE CHASSIS

Orchestrator / agent loop
Tool registry (API, DB, MCP)
Memory — short / long term
Execution sandbox

02

GUARDRAILS

≈ TRACK LIMITS & WALLS

Policy engine (safety)
Permissions — allow / deny
Structured-output constraints
Validation (schema, types)

03

OPERATIONAL LIMITS

≈ FUEL & BRAKES

Max tool calls / step
Max context window
Timeouts · rate limits
Compute / cost budget

04

TELEMETRY

≈ THE PIT-WALL FEED

LLM-call tracing
Tool-execution logs
Tokens · latency · cost
Eval signals (hallucination)

05

CONTROLLER LOOP

≈ RACE STRATEGY

Planner / router
Reflection / retry logic
Fallback strategies
Human-in-the-loop approval

RACE KEY · WHAT MAPS TO WHAT

DRIVER = LLM

reasons, decides, plans

CAR = AGENTIC RUNTIME

executes, uses tools, acts

FUEL = TOKENS

context & compute burned per lap — metered, budgeted

FIA · REGULATIONS

policy · safety · compliance

OBJECTIVE

real performance, safely — every lap better

04 / THE PIT CREW — NEW ROLES ATTACHING TO PRODUCTION

YESTERDAY'S CREW

SRE · DevOps

kept the running system alive

HARNESS ENGINEER

≈ CHASSIS & MECHANICS

Builds the machine — tools, sandboxes, guardrails, orchestration, observability.

FORWARD DEPLOYED ENGINEER

≈ RACE ENGINEER · TRACKSIDE

Embedded with users — tunes the harness to the real track, in real time.

VIBE CODER

≈ TEST DRIVER

Drives on intent — fast laps from natural language, feeding what works back to the build.

The model has talent. The harness makes it a champion.

AGENTIC HARNESS · AH-001 · REV 1.1
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